S4 Appendix. Computational materials, software resources, and reporting conventions

Scope

This appendix consolidates the computational materials-and-methods details that are most useful for reproducibility review and for detector-style reporting checks such as SciScore. The main article retains the short study-design statements and availability language needed for the simulator-first narrative. S2 Appendix retains simulator architecture and canonical protocol framing. S3 Appendix retains the exploratory open-data bridge pipelines. This appendix collects software identifiers, computational study-design conventions, stochastic-replication policy, and reproducibility entry points in one place.

Code and dataset identifiers

Resource	Type	Identifier	Role in this study
`cytodendaccessmodel` GitHub repository	code	NoeticDiffusion / cytodendaccessmodel	public source repository for simulator, analysis scripts, figures, manuscript files, and reviewer routing
`cytodendaccessmodel` Zenodo snapshot	archived code	DOI / 10.5281 / zenodo.20813821	earlier manuscript code release; predates E025 and requires a new version containing the E025 source and outputs
E025 preregistration and manifests	protocol	E025 registry files	frozen tuning plan, parameter registry, topology manifest, and probe manifest
E025 runner and renderer	code	E025 Python scripts	holdout analysis and reproducible publication-facing comparison figure
E025 holdout outputs	simulator data	E025 result folder	candidate scores, draw-level outputs, corrected comparisons, hub sensitivity, and hash manifest
DANDI dataset 000718	dataset	Dandiset / 000718	supplementary offline enrichment and replay-linked reuse bridge in S3
DANDI dataset 000336	dataset	Dandiset / 000336	supplementary cross-plane coupling bridge in S3
DANDI dataset 001710	dataset	Dandiset / 001710	supplementary perturbation-sensitive stabilization bridge in S3
Reviewer notebook	reviewer guide	root notebook	top-level reviewer routing for article evidence tiers, legacy lineage, and supplementary bridges
Claim map	claim map	root markdown	maps article claims and figures to experiments, result directories, and reviewer routing

Table S4.1: Primary code, data, and reviewer-routing identifiers for the simulator-first manuscript.

Software environment and identifiers

Unless otherwise noted, the local manuscript build environment used:

Software	Version	Identifier	Role
Python	`3.12.0`	`RRID:SCR_008394`	main implementation and analysis language; project requirement is `>=3.10`
NumPy	`1.26.4`	`RRID:SCR_008633`	array and numerical operations in simulator and analysis pipelines
PyYAML	`6.0.2`	no RRID listed here	dataset-selection and configuration parsing
PyTorch	`2.5.1+cu121`	no RRID listed here	declared project dependency for computational workflows
DANDI Python package / CLI	`0.74.3`	`RRID:SCR_017571`	archive listing, download, and metadata workflow for supplementary open-data analyses
PyNWB	`3.1.3`	`RRID:SCR_017452`	reading NWB files for the supplementary DANDI pipelines
matplotlib	`3.9.2`	no RRID listed here	article and analysis figure generation
pytest	`9.0.3`	no RRID listed here	test-suite execution
Typst	`0.14.1`	no RRID listed here	manuscript and appendix compilation

Table S4.2: Software environment and identifiers used for the local manuscript build. RRIDs are included where a validated software identifier was available during article preparation.

Computational study design

The primary study is a simulator-first model-discrimination analysis rather than a subject-sampling experiment. Its evidence includes historical mechanism and robustness experiments plus the E025 tuned model comparison:

- `E017`: canonical traceable simulator run and locked SIG-A through SIG-E reference
- `E018`: baseline comparator matrix
- `E019`: one-at-a-time robustness
- `E020`: paired-parameter robustness heatmaps
- `E021` and `E021R`: branch-count scaling, motif generalization, and specificity boundaries
- `E022` and `E022R`: hard comparators and shuffled-replay audit
- `E024`: historical common-probe protocol-development run; its estimates are not pooled with E025
- `E025`: equal-budget candidate tuning, validation, and locked holdout comparison across ten model families

Legacy experiments `exp001--exp016`, `exp013_paper_summary.py`, and `exp_seed_validation.py` are retained for lineage and reviewer cross-checks but are not the main evidence ladder for the current manuscript.

Canonical simulator protocol summary

The canonical simulator used by `E017` and `E018` contains four branches and two traces, with one shared overlap branch and private branches for each trace. The canonical parameterization matches the locked reference stack:

- `structural_lr = 0.18`
- `replay_gain = 0.80`
- `eligibility_decay = 0.12`
- `structural_decay = 0.005`

- `structural_gain = 6.0`
- `readout_threshold = 0.3`

The canonical protocol consists of:

1. two cue passes for the first trace,
2. two cue passes for the second trace,
3. pre-consolidation probes,
4. nine replay-dependent consolidation passes,
5. overlap-branch damage plus nine null passes,
6. targeted rescue with three rescue rounds and post-rescue probes.

Branch-level trace export includes fast activity, fast access, slow access, effective access, eligibility, consolidation support, structural accessibility, replay/input drive, and linking contributions.

Comparator implementation routing

Two comparator families were used in the primary evidence stack.

Comparator	Family	Implementation note
`fast_context_only`	baseline	fast gating retained; `M_b` frozen
`replay_no_structure`	baseline	replay updates transient variables but cannot write `M_b`
`random_slow_drift`	baseline	matched-magnitude slow drift replaces replay-specific writing
`fixed_allocation_only`	baseline	static overlap geometry without dynamic update
`hebbian_weight_only`	hard	weight-only alternative without overlap-specific slow writing
`soma_global_gain_only`	hard	uniform branch-independent gain producing non-specific recruitment
`shuffled_replay`	hard	replay preserved but branch identity shuffled
`eligibility_only`	hard	transient eligibility without retained structure
`resource_only`	hard	replay-linked resources without retained `M_b`

Table S4-3. Comparator routing for the simulator-first manuscript. Baseline comparators are formalized in E018; hard comparators are added in E022.

Robustness, scaling, and specificity design

The robustness and extension experiments are organized as follows:

- `E019` varies nine parameter families one at a time. Only the structural-noise sweep is stochastic; it uses `10` predeclared seeds (`0`, `1`, `2`, `3`, `4`, `42`, `101`, `202`, `303`, `404`). Non-noisy sweeps use the deterministic default seed.
- `E020` evaluates six paired-parameter heatmaps. The `structural_noise × replay_gain` grid uses the same `10` predeclared seeds per noisy cell; the remaining grids are deterministic.
- `E021` evaluates canonical, weak-overlap, strong-overlap, chain-overlap, hub-overlap, and sparse-random motifs across branch counts from `4` to `32`. Sparse-random motifs use seeds `42` and `123`.
- `E021R` converts those motif outcomes into reviewer-facing specificity classes such as expected weak-overlap failure and hub overlinking boundary.

- `E022R` evaluates shuffled replay with `20` shuffled seeds per `(motif_type, n_branches)` condition, with the full-model reference held at a fixed base seed.

Replication, randomness, and exclusions

The canonical reference experiments are deterministic once the protocol is fixed. The stochastic experiments are technical robustness replicates of the implemented simulator, not biological replicates. Seed lists are predeclared in code rather than tuned after inspection of the outcomes.

No outcome-driven exclusions are used in the primary simulator evidence ladder. Invalid or missing outputs are intended to surface as failures that require inspection rather than being silently imputed into the summary tables. Open-data QC rules, partial-day coverage handling, and dataset-specific exclusions are reported separately in S3 Appendix because they belong to the exploratory bridge tier rather than to the primary simulator evidence ladder.

Reporting conventions and not-applicable items

The simulator-first article does not report new human-participant recruitment, live animal experimentation by the author, cell-line work, antibody use, surgery, or specimen collection. No investigator blinding was performed for the simulator experiments because conditions are defined and labeled in automated scripts. Subject randomization is not applicable to the primary simulator experiments. A formal power analysis was not used to choose the deterministic canonical reference runs; stochastic seed counts were chosen as robustness and specificity audits.

Experimental subjects were not randomized into groups because the primary evidence ladder is a computational simulator study and does not assign biological subjects to intervention groups. We did not check for sample sizes using a power analysis because the primary simulator evidence ladder is based on deterministic reference runs plus predeclared robustness audits rather than on biological subject sampling. No simulator runs in the primary evidence ladder were excluded on the basis of their outcomes.

For the supplementary DANDI analyses, the relevant public archive identifiers, null constructions, and QC rules are documented in S3 Appendix and in the original archive records.

Reproducibility entry points

Reviewers who want the shortest route through the current manuscript should begin with:

- `project/neural_computation/e025_preregistration_plan_v1_0_frozen_20260925.md` (byte-exact frozen plan; SHA-256 in the pre-run freeze)
- `project/neural_computation/e025_preregistration_plan.md` (working copy with retrospective notes)
- `project/neural_computation/e025_inference_sensitivity_20261030.md` and `experiments/exp025_cluster_bootstrap_sensitivity.py` (post-hoc sensitivity analysis)
- `results/e025_common_probe_tuning/sensitivity/centered_cluster_bootstrap_holm.csv`
- `project/neural_computation/e025_parameter_registry.csv`
- `project/neural_computation/e025_topology_manifest.csv` and `e025_probe_manifest.csv`
- `experiments/exp025_tuned_common_probe.py`
- `results/e025_common_probe_tuning/summary/holdout_comparisons.csv`
- `results/e025_common_probe_tuning/scores/holdout_draw_scores.csv`
- `reviewer_slow_branch_level_accessibility.ipynb`
- `CLAIMS_TO_EXPERIMENTS.md`
- `results/e017_traceable_simulator_core/`

- `results/e018_comparator_trace_matrix/`
- `results/e019_one_at_a_time_parameter_robustness/`
- `results/e020_two_parameter_robustness_heatmaps/`
- `results/e021_scaling_and_motif_generalization/`
- `results/e021r_generalized_specificity_gate/`
- `results/e022_hard_comparators/`
- `results/e022r_shuffled_replay_scaling_audit/`

This routing keeps the current article's main simulator evidence, older lineage experiments, and supplementary open-data bridges explicitly separated.

Bibliography